ABSTRACT

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Title : STEP File Tokenizer for Transformer-based Annotation

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Current CAD asset management is hindered by manual annotation and automated methods relying on lossy visual conversions or unavailable construction histories. This study proposes a novel Transformer-based architecture for the direct hierarchical annotation of raw Boundary Representation (B-Rep) STEP files. We introduce a Hierarchical Dual-Stream Tokenizer utilizing Semantic Macro-Compression to reduce sequence complexity, Canonical Normalization via PCA for rotational and translational invariance, and deterministic Topological Linearization.Leveraging a Qwen 3.5B backbone, the architecture employs Dual-Stream Encoding to fuse discrete semantic tokens with complex-valued geometric embeddings via Geometry-Infused Attention and Rotary Positional Embeddings (RoPE). Following the LIMA principle, the model was fine-tuned on $N = 2,500$ high-fidelity, human-verified CAD files.Results demonstrate robust performance across a three-tiered schema: Level 1 functional classification (BERTScore: 0.95), Level 2 structural description (ROUGE-L: 0.88), and Level 3 parametric precision via constrained decoding (BLEU-4: 0.85). Inter-rater reliability was successfully validated using Cohen's Kappa. This scalable framework effectively captures functional identity and precise parametric constraints without requiring procedural histories, revolutionizing industrial CAD management.

Keywords: CAD, STEP, B-Rep, Transformer, Dual-Stream Tokenizer, Semantic Macro-Compression, PCA, RoPE, BERTScore, LIMA, inter-rater reliability, ROUGE-L, BLEU, Cohen's Kappa.